



The University of Lahore (UOL), Lahore

Course Name:	Operating Systems (Theory)	Course Code:	CS11303 11	Credit Hours:	3+1
Course Instructor/s:	Muhammad Afnan	Program Name	BS Software Engineering		
Semester/Sections:	BSSE – 6S, 6T	Assignment No	4		
Submission Deadline:	Thursday, May 21, 2026 Till 11:59 PM	Maximum Marks:	10		

Instructions:

- Understanding the questions is part of the assignment.
- *This is a handwritten assignment. Typed/MS Word submissions will not be accepted or graded.*
- Use clear A4 size page and original wording (plagiarism = 0 marks). Zero tolerance for plagiarism, copied work will be marked ZERO.
- Write your Name, Roll Number, Section, and Due Date on the first page.
- Submit your assignment in PDF format on the LMS/SAP portal within the given time frame. (Emailed or late submissions will not be accepted.)

Assignment 01

Question 1:

Consider the following reference string of page numbers:
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

Assume **3 frames** are available in memory.

Tasks:

Part	Task
(a)	Simulate FIFO algorithm. Show frames after each reference. Calculate total page faults.
(b)	Simulate LRU algorithm. Show frames after each reference. Calculate total page faults.
(c)	Simulate OPT algorithm. Show frames after each reference. Calculate total page faults.
(d)	Draw a block diagram showing the replacement process when page 4 is referenced in FIFO.
(e)	Compare results and state which algorithm performs best and why it cannot be implemented.

